

Note-A review of vorticity conditions in the numerical solution

Information

- Article: A review of vorticity conditions in the numerical solution of the $\zeta - \psi$ equations
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Summary of the content

In this article, the authors present and compare different numerical scheme to determine the correct the vorticity boundary values, including finite difference, finite elements, coupled formulations (local formula of boundary vorticity), and uncoupled formulations (based on the integral constraint, i.e Quartapelle's projection condition).

Methodology

- For one-dimensional case: Compare the numerical results of different methods. The values for accuracy estimation are

$$\text{global relative error: } g.r.e(u) = \frac{\|u^c - u^e\|}{\|u^e\|}$$

$$\text{local relative error: } l.r.e(u_i) = \frac{|u_i^c - u_i^e|}{|u_i^e|}, \quad i = 0 \quad \text{or} \quad i = N$$

$$\text{order of accuracy: } p = \frac{\ln(r.e._H / r.e._h)}{\ln(H/h)}$$

where $u^c(x)$ is the computed solution, and $u^e(x)$ is the exact solution. $\|u\|$ denotes the L^1 norm of u . H and h are two mesh sizes, with $H > h$.

- For two-dimensional case: The benchmark for the two-dimensional case is *Lid-Driven Cavity of Stokes Flow*. The global relative error is defined in a similar way as the one-dimensional case, meaning

$$g.r.e(u) = \frac{\|u^c - u^e\|}{\|u^e\|}$$

where $||u||$ is the L^1 norm of u defined as

$$||u|| = \int_0^1 \int_0^1 |u(x, y)| dx dy$$

Here are some important quotes from the journal.

Quote (P140.):

As a consequence, the vorticity boundary formula can be used in two basically different manners: (1) in coupled solution methods, where the discrete equations are solved as a global system of equations comprising both the vorticity and the stream function unknowns; (2) in uncoupled iterative solution procedures, where the values of vorticity provided by the boundary formula are used as Dirichlet data for the vorticity equation, as done, e.g., in the early solutions of the vorticity and stream function equations by Allen and Southwell .

Quote (About the equivalence of uncoupled method and coupled method, P141.):

The three uncoupled methods are shown to be **algebraically equivalent and numerically identical** to the coupled method based on the Thom/ Burggraf formula. A version of the influence matrix method incorporating Woods' formula is finally elaborated upon and shown to provide identical results to the coupled method using the same boundary treatment. *Then, a two-dimensional model (Stokes) problem and the classical driven cavity flow [16] are considered to extend the validity of the previous 1D results to the case of 2D problems.*

Quote (P142.):

A basic difficulty in solving the system of second-order elliptic equations for ζ and ψ supplemented by the boundary conditions above is that two boundary conditions are specified for ψ while none is available for ζ . This occurrence does not mean that the problem for ψ is overdetermined but only that ζ , to be the solution to the problem, must be compatible in some way with the prescribed boundary values a and b . There are two basically different ways for achieving such a 'coherence' between ζ and the prescribed velocity boundary values: (1) solving the system of coupled equations, by means of either direct or iterative methods; (2) uncoupling the equations by introducing appropriate conditions for the vorticity which are the mathematical equivalent of the boundary conditions for the stream function.

Remark (P164-166)

From the results given in the Tables it appears that, like in the one-dimensional case: (1) all results are second-order-accurate for sufficiently large values of N ; (2) the Thom/Burggraf formula, having the same order of accuracy as the interior points of discretization, achieves the second-order accuracy sooner and more smoothly in the overall domain (Tables 7-10), much later and with some difficulty at boundary points (Table 11); (3) higher-order formulas are all essentially equivalent and provide errors at least five times smaller than those obtained using the

Thom/Burggraf formula; (4) *last, but not least, the coupled method using the Thom/Burggraf formula and the uncoupled methods provide identical results, within round-off error. Furthermore, only these solutions satisfy the discrete counterpart of the global constraint:*

$$\int \int \zeta dV = - \oint b ds$$

within round-off error, the surface and line integrals being evaluated by means of the standard trapezoidal rules, see, e.g., Eqs. (95) and (86), respectively.

Remark (P167-168)

It appears that also for the two-dimensional nonlinear Navier-Stokes equations, like for the previous model problem: (1) all results appear to become second-order accurate for sufficiently large values of N , N increasing with Re , as obvious; (2) the Thom/Burggraf formula achieves the theoretical order of accuracy more smoothly than the higherorder formulas; (3) the superior accuracy of higher-order formulas with respect to the Thom/ Burggraf one is significantly reduced, except for the two coarsest grids, by the presence of the nonlinear term; (4) *last, but not least, uncoupled methods are again algebraically identical to the coupled method using the Thom/Burggraf formula. Most importantly, only these two (identical) solutions satisfy the ‘conservation law’ :*

$$\int \int \zeta dV = - \oint b ds = -1$$

What do I learn/understand ?

- The local formulas of boundary vorticity should be only considered as a part of the Taylor expansion of the stream-function ψ near the boundary. In the most general case, the uncoupled methods have the clearest physical/mathematical meaning because they impose the vorticity integral constraints directly. Coupled/local-formula methods may inherit the same meaning only when they are algebraically equivalent to those integral constraints. However, if the those equivalences are violated by some effects, such as the geometry of domain, mesh size or type, there is no guarantee that coupled methods can provide the correct result, because the global constraint represents the approximation of distribution of vorticity in the domain. (This thought is related to the article: P. M. Gresho, 1992)
- For the tests with Lid-Driven Cavity of steady Stokes/Navier-Stokes flow, there is an algebraic equivalence and numerical equivalence between the uncoupled methods and coupled methods which means we can get the same vorticity boundary value from those methods. The validity still hold for the Navier-Stokes equation.

Reference

- L. Quartapelle and F. Valz-Gris, Projection conditions on the vorticity in viscous incompressible flows, *Int. J. Numer. Meth. Fluids* 1 (1981) 129–144.
- Weinan E, Jian-Guo Liu, Vorticity Boundary Condition and Related Issues for Finite Difference Schemes, *Journal of Computational Physics* Volume 124, Issue 2, 15 March 1996, Pages 368-382
- P.M. Gresho, Some interesting issues in incompressible fluid dynamics, both in the continuum and in numerical simulation, *Advances in Applied Mechanics*, 28 (1992)